

DeepSeek-V2

A Strong, Economical, and Efficient MoE Language Model

236B

Total Params

21B

Activated Params

128K

Context Length

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

Larger Models = Better Intelligence

Scaling up parameter count is the most reliable path to stronger reasoning.

The Prohibitive Costs of Dense Scaling

- Dense training requires astronomical compute budgets.
- Inference is bottlenecked by the Multi-Head Attention (MHA) KV Cache.

The MoE + Efficient Attention Imperative

To overcome these barriers, DeepSeek-V2 combines a novel MoE structure with compressed attention.

DeepSeek-V2 Architecture

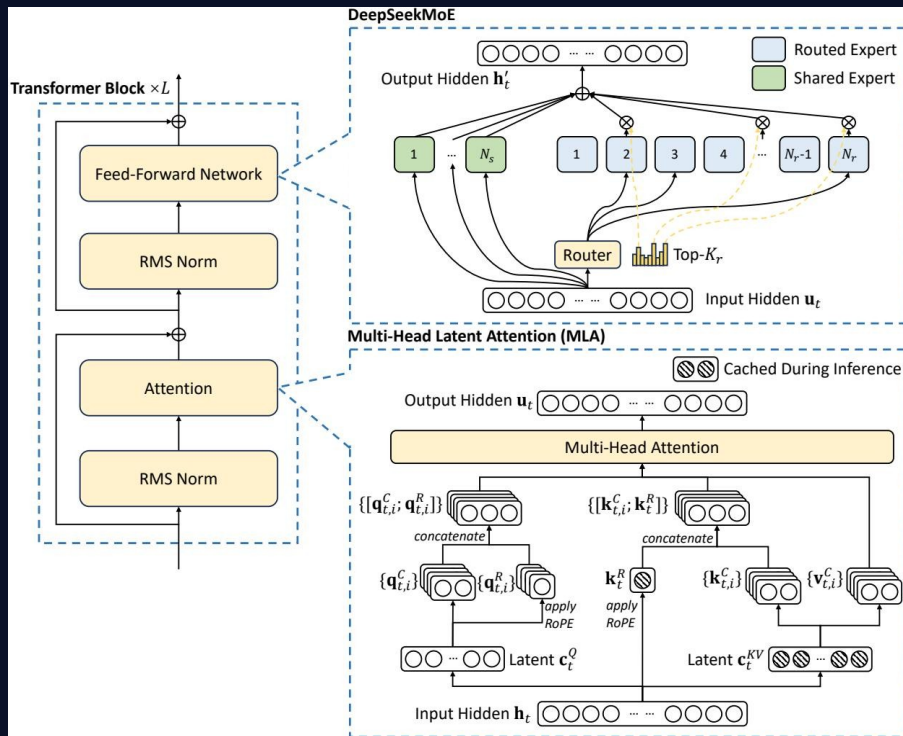
MLA + DeepSeekMoE in a Transformer Block

Multi-Head Latent Attention (MLA)

Replaces standard MHA for efficient KV caching.

DeepSeekMoE Feed-Forward Network

Replaces dense FFN with fine-grained routed & shared experts.



Multi-Head Latent Attention (MLA)

Compresses KV Cache by 93.3%

Low-Rank Key-Value Joint Compression

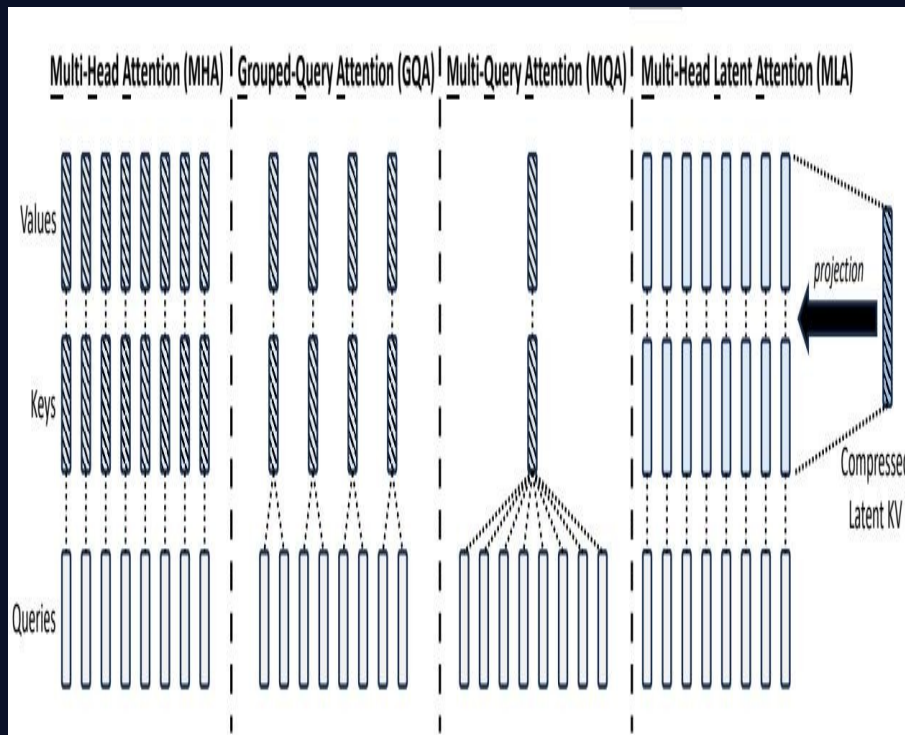
Instead of caching full K and V vectors per head, MLA compresses them into a single compact latent vector.

Decoupled RoPE Positional Encoding

Positional keys are decoupled and cached efficiently.

Strictly Superior to GQA/MQA

Better performance than full MHA while needing drastically less memory.



DeepSeekMoE: Fine-Grained Experts

Shared Expert Isolation and Device-Limited Routing

Shared Experts

Always activated to capture common, generalized knowledge.

Fine-Grained Routed Experts

Sparsely activated via a top-K router for specialized domains.

Device-Limited Routing

Constrains tokens to experts on limited devices, minimizing cross-node communication overhead.

Auxiliary Load Balancing

Enforces workload balance to prevent expert overflow.

Unprecedented Efficiency Gains

DeepSeek-V2 vs. DeepSeek 67B

Training Cost

-42.5%

Compared to 67B Dense

KV Cache Size

-93.3%

Via MLA Compression

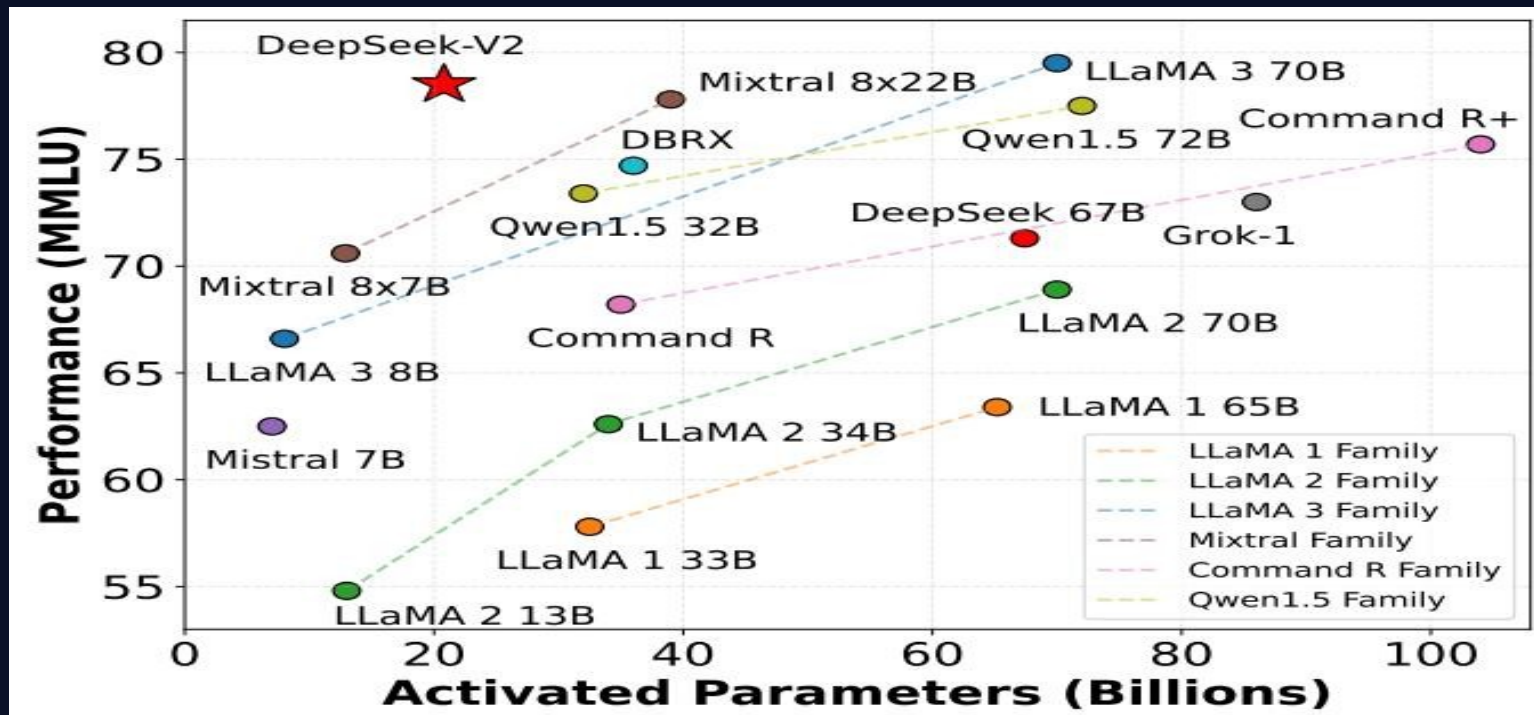
Max Throughput

5.76×

Generation Speed

Pareto-Optimal Performance

DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters



Strong Multilingual Alignment

DeepSeek-V2 Chat (RL) Benchmarks

Alignment Pipeline

1. Supervised Fine-Tuning (SFT) on 1.5M conversational sessions.
2. Reinforcement Learning via Group Relative Policy Optimization (GRPO).

Benchmark	DeepSeek-V2 Chat (RL)
AlpacaEval 2.0 (length-controlled win rate)	38.9%
MT-Bench (overall)	8.97
AlignBench (overall, Chinese)	7.91

Outperforms all open-source models and most closed-source models in Chinese.

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

MLA Solves the KV Cache Bottleneck

Reduces cache memory by 93.3% and acts as a strictly superior alternative to GQA/MQA for inference efficiency.

DeepSeekMoE Drives Economical Training

Shared expert isolation and fine-grained experts cut 236B training costs by 42.5% compared to 67B dense models.

Pareto-Optimal Performance

Reaches 78.5 MMLU using only 21B activated parameters, dominating 70B-class dense models in efficiency.

Unlocks Large-Scale Deployment

A 5.76× generation throughput boost makes frontier-class reasoning practical to serve in production environments.